



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE
NASIONALE
SENIOR SERTIFIKAAT**

GRADE 12/GRAAD 12

TECHNICAL MATHEMATICS P1/TEGNIJSE WISKUNDE V1

NOVEMBER 2024

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

MARKING CODES/NASIENKODES	
A	Accuracy/Akkuraatheid
CA	Consistent accuracy/Volgehoe akkuraatheid
M	Method/Metode
R	Rounding/Afronding
NPR	No penalty for rounding/Geen penalisering vir afronding nie
NPU	No penalty for units omitted/Geen penalisering vir eenhede weggelaat nie
S	Simplification/Vereenvoudiging
SF	Substitution in correct formula/Vervanging in korrekte formule

**These marking guidelines consist of 19 pages.
Hierdie nasienriglyne bestaan uit 19 bladsye.**

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- Consistent accuracy (CA) applies in all aspects of the marking guidelines where indicated.
- No penalty for rounding (NPR) for ALL questions.
- # Shows questions where a Tolerance Range will be applied:
Q 2.2 ; Q 4.1.5 ; Q 5.4 & Q 9.2.2

LET WEL:

- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, sien slegs die EERSTE poging na.
- Volgehoue akkuraatheid (CA) is deurgaans op alle aspekte van die nasienriglyne van toepassing soos aangedui.
- Geen penaliserings vir afronding (NPR) vir ALLE vrae nie.
- # Toon vrae waar Toleransie wydte (Verdraagsaamheids-omvang) toegepas word: **V 2.2 ; V 4.1.5 ; V 5.4 & V 9.2.2**

QUESTION/VRAAG 1

1.1.1	$x(2x + 7) = 0$ $x = 0$ or / of $-\frac{7}{2}$ OR/OF $x = -3,5$	$\checkmark 0$ A $\checkmark -\frac{7}{2}$ OR/OF $-3,5$ A (2)
1.1.2	$3x^2 + x = 6 + 5x$ $3x^2 - 4x - 6 = 0$ OR/OF $0 = -3x^2 + 4x + 6$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)(-6)}}{2(3)}$ OR/OF $= \frac{-(4) \pm \sqrt{(4)^2 - 4(-3)(6)}}{2(-3)}$ $= \frac{4 \pm \sqrt{88}}{6}$ $\therefore x \approx 2,23$ or / of $x \approx -0,90$ Refer to the Addendum/ Verwys na die Addendum	$\checkmark$ std form/vorm A $\checkmark$ SF CA $\checkmark x \approx 2,23$ CA $\checkmark x \approx -0,90$ CA NPR (4)

1.1.3	$x^2 + 3x - 10 \leq 0$ $(x - 2)(x + 5) \leq 0$ OR/OF $\frac{-(3) \pm \sqrt{(3)^2 - 4(1)(-10)}}{2(1)}$ Critical values/Kritieke waardes: 2 and/en -5 $\therefore -5 \leq x \leq 2$ OR/OF $x \in [-5; 2]$ OR/OF $x \geq -5$ and/en $x \leq 2$ OR/OF  OR/OF Accept: From -5 to 2 / Aanvaar: Vanaf -5 tot 2	✓ factors/faktore/SF A ✓ both critical values/ kritieke waardes CA ✓ correct notation/ korrekte notasie/ correct graphical solution/korrekte grafiese oplossing A AO: Full Marks/ Volpunte (3)
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1.2	$y - x = 2 \text{ and/en } x^2 + y^2 = 20$ $x = y - 2$ $(y - 2)^2 + y^2 = 20$ $y^2 - 4y + 4 + y^2 = 20$ $2y^2 - 4y - 16 = 0$ $2(y - 4)(y + 2) = 0 \text{ OR/OF } y = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(2)(-16)}}{2(2)}$ $\therefore y = 4 \text{ or/of } y = -2$ $\therefore x = 4 - 2 = 2 \text{ or/of } x = -2 - 2 = -4$ OR/OF $y = x + 2$ $x^2 + (x + 2)^2 = 20$ $x^2 + x^2 + 4x + 4 = 20$ $2x^2 + 4x - 16 = 0$ $2(x - 2)(x + 4) = 0 \text{ OR/OF } x = \frac{-(4) \pm \sqrt{(4)^2 - 4(2)(-16)}}{2(2)}$ $\therefore x = 2 \text{ or/of } x = -4$ $\therefore y = 2 + 2 = 4 \text{ or/of } y = -4 + 2 = -2$ Refer to the Addendum/ Verwys na die Addendum	$\checkmark$ subject/onderwerp A $\checkmark$ subst./vervang CA $\checkmark$ std form/vorm CA $\checkmark$ Factors/Faktore/SF CA $\checkmark$ both y-values/beide y-wrdes CA $\checkmark$ both x-values/beide x-wrdes CA OR/OF $\checkmark$ subject/onderwerp A $\checkmark$ subst./vervang CA $\checkmark$ std form/vorm CA $\checkmark$ Factors/Faktore/SF CA $\checkmark$ both x-values/beide x-wrdes CA $\checkmark$ both y-values/beide y-wrdes CA (6)
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1.3.1	$CR = \frac{CV + SV}{SV}$ $SV \times CR = CV + SV$ $CV = SV \times CR - SV \quad \text{OR/OF} \quad CV = SV \times (CR - 1)$ OR/OF $CR = \frac{CV}{SV} + \frac{SV}{SV}$ $CR = \frac{CV}{SV} + 1$ $CV = SV \times (CR - 1)$	✓ multiplying by SV/ vermenigvuldiging met SV A ✓ CV subject/onderwerp CA OR/OF ✓ dividing/ verdeling A ✓ CV subject/ onderwerp CA (2)
1.3.2	$CV = SV \times CR - SV$ $= 48 \text{ cm}^3 \times \frac{9,5}{1} - 48 \text{ cm}^3$ $= 408 \text{ cm}^3$ OR/OF $CV = SV (CR - 1)$ $= 48 \text{ cm}^3 \left(\frac{9,5}{1} - 1 \right)$ $= 408 \text{ cm}^3$ OR/OF $CR = \frac{CV + SV}{SV}$ $\frac{9,5}{1} = \frac{CV + 48 \text{ cm}^3}{48 \text{ cm}^3}$ $\therefore CV = 408 \text{ cm}^3$	✓ SF CA ✓ S CA OR/OF ✓ SF CA ✓ S CA OR/OF ✓ SF A ✓ S NPU CA (2)

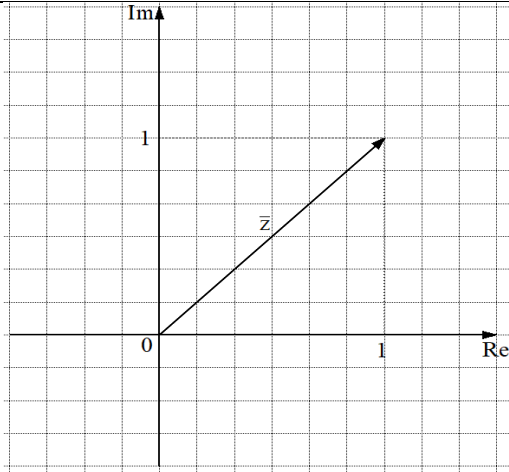
1.4	$1110_2 = 14$	✓14	A (1)
1.5	$1110_2 \times 35$ $= 14 \times 35 = 490 = 111101010_2$ OR/OF $1110_2 \times 100011_2 = 111101010_2$ Refer to the Addendum /Verwys na die Addendum	✓490 ✓ 111101010_2 OR/OF ✓ 100011_2 ✓ 111101010_2 AO: Full marks/Volpunte	CA CA A CA (2)
			[22]

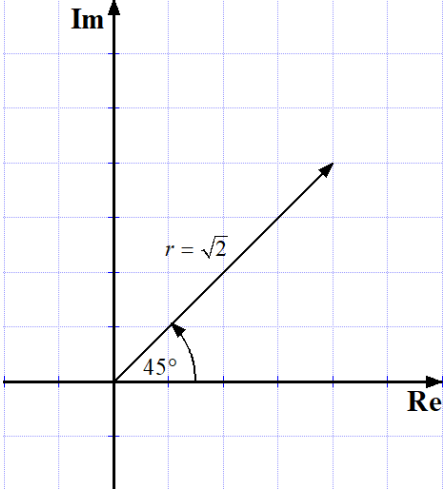
QUESTION/VRAAG 2

2.1.1	$p = 3$	✓ 3	A (1)
2.1.2	$1 - 7p < 0$ $p > \frac{1}{7}$ OR/OF $p > 0,14$	✓ $\Delta < 0$ ✓ $p > \frac{1}{7}$ OR/OF $p > 0,14$ AO: Full marks/Volpunte	A CA (2)
2.2	$3(x + 1) = x^2 + t$ $x^2 - 3x - 3 + t = 0$ OR/OF $-x^2 + 3x + 3 - t = 0$ $(-3)^2 - 4(1)(-3 + t) \geq 0$ OR/OF $(3)^2 - 4(-1)(3 - t) \geq 0$ $9 + 12 - 4t \geq 0$ $-4t \geq -21$ $\therefore t \leq \frac{21}{4}$ OR/OF 5,25	✓ standard form / <i>standaardvorm</i> A ✓ SF CA ✓ $\Delta \geq 0$ A ✓ value(s) of/ <i>waardes van</i> t CA	 (4)
			[7]

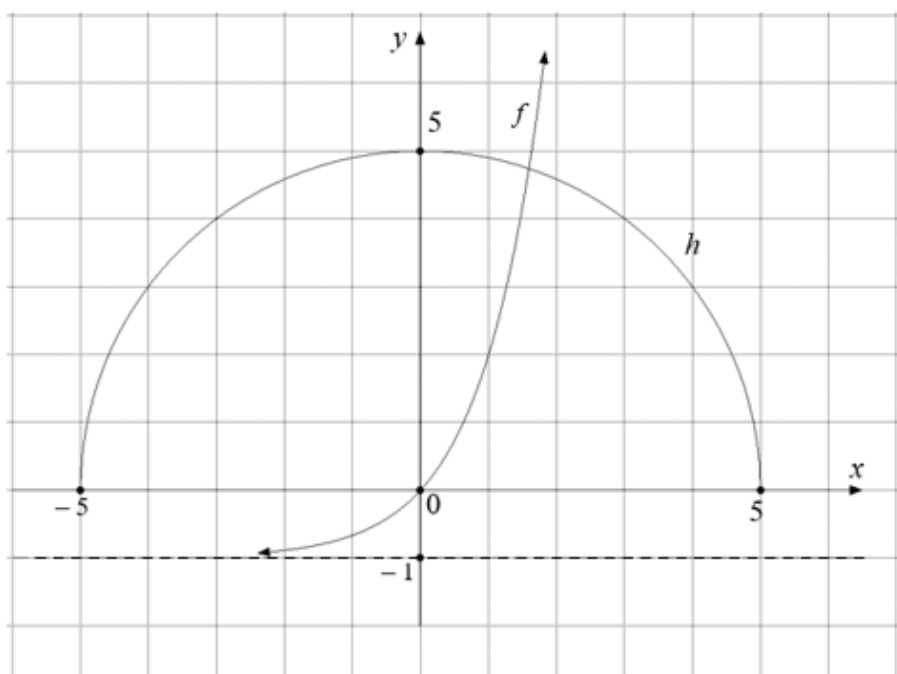
QUESTION/VRAAG 3

3.1.1	$27^{\frac{2}{3}} = 9$ OR/OF 3^2	✓ 9 OR/OF 3^2 A (1)
3.1.2	$(1 + \sqrt{3})^2 - \sqrt{12}$ $= 1 + 2\sqrt{3} + 3 - 2\sqrt{3}$ $= 4$ Refer to the Addendum/ Verwys na die Addendum	✓ expanding/ uitbreiding A ✓ $2\sqrt{3}$ A ✓ S CA AO : 1 mark/ punt (3)
3.1.3	$\log_p p = 1$	✓ 1 A (1)
3.1.4	$\log_3 81 - \log_2 \sin 30^\circ - \log_5 \sqrt{5}$ $= \log_3 3^4 - \log_2 \frac{1}{2} - \log_5 5^{\frac{1}{2}}$ OR/OF $= \frac{\log 3^4}{\log 3} - \frac{\log 2^{-1}}{\log 2} - \frac{\log 5^{\frac{1}{2}}}{\log 5}$ $= 4 + 1 - \frac{1}{2}$ $= 4\frac{1}{2}$ OR/OF $\frac{9}{2}$ OR/OF 4,5	✓ 4 A ✓ + 1 A ✓ $-\frac{1}{2}$ A ✓ S CA AO : 1 mark/ punt (4)

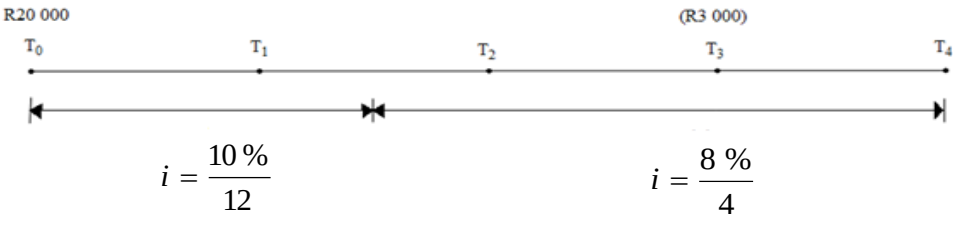
3.2	$5^{x+2} - 5^x = \frac{24}{5}$ $5^x \times 5^2 - 5^x = \frac{24}{5} \quad \text{OR/OF} \quad 5^x \times 25 - 5^x = \frac{24}{5}$ $5^x (5^2 - 1) = \frac{24}{5} \quad \text{OR/OF} \quad 5^x (24) = \frac{24}{5}$ $5^x = \frac{1}{5} = 5^{-1}$ $\therefore x = -1$ OR/OF $5^{x+2} - 5^x = \frac{24}{5}$ $5^{x+3} - 5^{x+1} = 24$ $5^x \times 5^3 - 5^x \times 5^1 = 24 \quad \text{OR/OF} \quad 5^x \times 125 - 5^x \times 5 = 24$ $5^x (5^3 - 5) = 24 \quad \text{OR/OF} \quad 5^x (120) = 24$ $5^x = \frac{1}{5} = 5^{-1}$ $\therefore x = -1$	✓ exp.prop./eksp.einsk. A ✓ S CA ✓ value of/waarde van x CA OR/OF ✓ exp.prop./eksp.einsk. A ✓ S CA ✓ value of/waarde van x CA AO: Full marks/Volpunte (3)
3.3.1	$z_1 = 1 - i$	✓ form/vorm A (1)
3.3.2	$\bar{z}_1 = 1 + i$	✓ conjugate/konjugeerde CA (1)
3.3.3	 OR/OF	✓ Real part/Reële deel CA ✓ Imaginary part/ Imaginêre deel CA ✓ terminal arm or point where Re and Im intersect/terminale arm of punt waar Re en Im sny CA OR/OF

		✓ Modulus/ <i>Modulus</i> CA ✓✓ Argument/ <i>Argument</i> CA (3)
3.3.4	$r = \sqrt{1^2 + (-1)^2}$ $= \sqrt{2} \text{ OR/OF } \approx 1,41$ $\text{ref./verwys } \angle = \tan^{-1}\left(\frac{1}{1}\right)$ $\text{ref./verwys } \angle = 45^\circ$ $\therefore \theta = 360^\circ - 45^\circ = 315^\circ$ $\therefore z_1 = \sqrt{2} \text{ cis } 315^\circ \text{ OR/OF } z_1 \approx 1,41 \text{ cis } 315^\circ$ OR/OF $r = \sqrt{1^2 + (-1)^2}$ $= \sqrt{2} \text{ OR/OF } \approx 1,41$ $\theta = \tan^{-1}\left(-\frac{1}{1}\right)$ $\theta = -45^\circ$ $\therefore z_1 = \sqrt{2} \text{ cis } (-45^\circ) \text{ OR/OF } \therefore z_1 = 1,41 \text{ cis } (-45^\circ)$	✓SF CA ✓ <i>r</i> CA ✓ref./verwys $\angle$ CA ✓correct quadrant / <i>korrekte kwadrant</i> CA ✓polar form / <i>polêre vorm</i> CA OR/OF ✓SF CA ✓ <i>r</i> CA ✓✓ -45° CA ✓correct polar form / <i>korrekte polêre vorm</i> CA AO: Full marks/Volpunte (5)
		[22]

QUESTION 4/VRAAG 4

4.1.1	$y = -1$	✓ equation / vergelyking	A (1)
4.1.2	$-5 \leq x \leq 5$ OR/OF $x \in [-5; 5]$ OR/OF $x \leq 5$ and / en $x \geq -5$ OR/OF Accept: From -5 to 5 / Aanvaar: Vanaf -5 tot 5	✓ critical values / kritieke waardes ✓ notation / notasie	A A (2)
4.1.3	$0 = 3^x - 1$ $3^x = 1$ $3^x = 3^0$ $x = 0$ Accept/Aanvaar: $(0; 0)$	✓ subst / vervang. $y = 0$ ✓ $x = 0$ AO: Full marks/Volpunte	A A (2)
4.1.4	$y = 3^0 - 1$ $y = 0$ Accept/Aanvaar: $(0; 0)$	✓ subst. / vervang $x = 0$ ✓ $y = 0$ AO: Full marks/Volpunte	A CA (2)
4.1.5	 For/vir h: ✓ shape / vorm ✓ all intercepts on axes /alle afsnitte op die asse A A For/ vir f: ✓ shape / vorm ✓ asymptote / asimptoot ✓ intercept(s) / afsnitte A CA CA		

(5)

5.4		
	$A = 20\,000 \left(1 + \frac{10\%}{12}\right)^{1,5 \times 12} \approx R\,23\,222,25$ $\approx 23\,222,25 \left(1 + \frac{8\%}{4}\right)^{1,5 \times 4} \approx R\,26\,152,03$ $\therefore 26\,152,03 - 3\,000$ $\approx R\,23\,152,03$ $\approx 23\,152,03 \left(1 + \frac{8\%}{4}\right)^{1 \times 4}$ $\approx R\,25\,060,49$ OR/OF $A = 20\,000 \left(1 + \frac{10\%}{12}\right)^{1,5 \times 12} \left(1 + \frac{8\%}{4}\right)^{2,5 \times 4}$ $- 3\,000 \left(1 + \frac{8\%}{4}\right)^{1 \times 4}$ $\approx R\,25\,060,49$ OR/OF $A = \left[20\,000 \left(1 + \frac{10\%}{12}\right)^{1,5 \times 12} \left(1 + \frac{8\%}{4}\right)^{1,5 \times 4} - 3\,000 \right]$ $\times \left(1 + \frac{8\%}{4}\right)^{1 \times 4}$ $\approx R\,25\,060,49$	$\checkmark 20\,000 \left(1 + \frac{10\%}{12}\right)^{1,5 \times 12} \quad \mathbf{A}$ $\checkmark 23\,222,25 \left(1 + \frac{8\%}{4}\right)^{1,5 \times 4} \quad \mathbf{CA}$ $\checkmark \mathbf{M} \quad \mathbf{A}$ $\checkmark 23\,152,03 \left(1 + \frac{8\%}{4}\right)^{1 \times 4} \quad \mathbf{CA}$ $\checkmark \approx R\,25\,060,49 \quad \mathbf{CA}$ OR/OF $\checkmark 20\,000 \left(1 + \frac{10\%}{12}\right)^{1,5 \times 12} \quad \mathbf{A}$ $\checkmark \left(1 + \frac{8\%}{4}\right)^{2,5 \times 4} \quad \mathbf{A}$ $\checkmark \mathbf{M} \quad \mathbf{A}$ $\checkmark \left(1 + \frac{8\%}{4}\right)^{1 \times 4} \quad \mathbf{A}$ $\checkmark \approx R\,25\,060,49 \quad \mathbf{CA}$ OR/OF $\checkmark 20\,000 \left(1 + \frac{10\%}{12}\right)^{1,5 \times 12} \quad \mathbf{A}$ $\checkmark \left(1 + \frac{8\%}{4}\right)^{1,5 \times 4} \quad \mathbf{A}$ $\checkmark \mathbf{M} \quad \mathbf{A}$ $\checkmark \left(1 + \frac{8\%}{4}\right)^{1 \times 4} \quad \mathbf{A}$ $\checkmark \approx R\,25\,060,49 \quad \mathbf{CA}$ (5)
		[17]

QUESTION/VRAAG 6

6.1	$f(x) = 9x - 6$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $= \lim_{h \rightarrow 0} \frac{9(x+h) - 6 - (9x - 6)}{h}$ $= \lim_{h \rightarrow 0} \frac{9x + 9h - 6 - 9x + 6}{h}$ $= \lim_{h \rightarrow 0} \frac{9h}{h}$ $= \lim_{h \rightarrow 0} (9)$ $\therefore f'(x) = 9$	✓ definition/definisie A ✓ SF A ✓ S CA ✓ S CA ✓ 9 CA Penalty: 1 mark for incorrect notation/ Penaliseer : 1 punt vir foutiewe notasie AO: 1mark/punt (5)
6.2	$f(x) = 11\pi^2$ $f'(x) = 0$	✓ 0 A (1)
6.3.1	$y = x \left(3 + \frac{3}{x^5} \right)$ $= 3x + 3x^{-4}$	✓ $3x$ A ✓ $3x^{-4}$ or/of $\frac{3}{x^4}$ A (2)
6.3.2	$\frac{dy}{dx} = 3 - 12x^{-5}$	✓ 3 CA ✓ $-12x^{-5}$ or/of $-\frac{12}{x^5}$ CA (2)
6.4.1	$\sqrt[5]{x^8} = x^{\frac{8}{5}}$	✓ $x^{\frac{8}{5}}$ A (1)
6.4.2	$D_x \left[x^{\frac{8}{5}} - 5x^{12} \right]$ $= \frac{8}{5} x^{\frac{3}{5}} - 60x^{11}$	✓ $\frac{8}{5} x^{\frac{3}{5}}$ CA ✓ $-60x^{11}$ A (2)

6.5.1	$g(x) = -x^3 + 6x^2$ $g'(x) = -3x^2 + 12x$	✓ $-3x^2$ ✓ $12x$	A A (2)
6.5.2	$g'(-2) = -3(-2)^2 + 12(-2)$ $= -36$	✓ Sub. into derivative / vervang in afgeleide ✓ -36 AO: Full marks/Volpunte	CA CA (2)
6.5.3	$-3x^2 + 12x = -36$ $-3x^2 + 12x + 36 = 0$ $-3(x-6)(x+2) = 0$ OR/OF $x = \frac{-(12) \pm \sqrt{(12)^2 - 4(-3)(36)}}{2(-3)}$ $\therefore x = 6$ $y = g(6) = -(6)^3 + 6(6)^2 = 0$ The other point is /Die ander punt is $(6 ; 0)$	✓ equating / gelykstelling ✓ factors/formula/faktore ✓ x-coordinate of other point / x-koördinaat van ander punt ✓ y-coordinate of other point / y-koördinaat van ander punt	CA CA CA CA (4) [21]

QUESTION/VRAAG 7

7.1	OD = 60 units/ <i>eenhede</i>	✓length/ <i>lengte</i> A (1)
7.2	x - intercepts / <i>afsnitte</i> $y = 0$ $(x + 2)(x^2 - x - 30) = 0$ $(x + 2)(x - 6)(x + 5) = 0$ OR/OF $x + 2 = 0$ or/of $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-30)}}{2(1)}$ $\therefore x = -2$ or/of $x = 6$ or/of $x = -5$ $\therefore A(-5;0)$ and/en $\therefore C(6;0)$	✓quadratic factor/ <i>kwadr. faktor</i> A ✓factors/ <i>faktore</i> / SF CA ✓coordinates of/ <i>koördinate van</i> A CA ✓coordinates of/ <i>koördinate van</i> C CA AO: Full marks/Volpunte (4)
7.3	$f'(x) = 3x^2 + 2x - 32 = 0$ $x = \frac{-(2) \pm \sqrt{(2)^2 - 4(3)(-32)}}{2(3)}$ $\therefore x \approx -3,62$ or/of $x \approx 2,95$ $f(2,95) = (2,95)^3 + (2,95)^2 - 32(2,95) - 60 \approx -120,03$ $\therefore G(-3,62; -120,03)$	✓derivative/ <i>afgeleide</i> A ✓equating derivative to 0/ <i>stel afgeleide gelyk aan 0</i> A ✓ SF CA ✓both values of / <i>beide waardes van</i> x CA ✓correct value of / <i>korrekte waarde van</i> y CA ✓coordinates of/ <i>koördinate van</i> G CA NPR (6)
7.4.1	$-5 \leq x \leq -2$ OR/OF $x \in [-5; -2]$ OR/OF $x \geq -5$ and/en $x \leq -2$ Accept: From -5 to -2 / <i>Aanvaar: Vanaf -5 tot -2</i>	✓critical values/ <i>kritieke waardes</i> CA ✓correct notation/ <i>korrekte notasie</i> A (2)
7.4.2	$-3,62 < x < 2,95$ OR/OF $x \in (-3,62; 2,95)$ OR/OF $x > -3,62$ and/en $x < 2,95$ Accept: Between $-3,62$ and $2,95$ / <i>Aanvaar: Tussen $-3,62$ en $2,95$</i>	✓critical values/ <i>kritieke waardes</i> CA ✓correct notation/ <i>korrekte notasie</i> A NPR (2)
		[15]

QUESTION/VRAAG 8

8.1	R10 000 Refer to the Addendum/ Verwys na die Addendum	✓R10 000 A (1)
8.2	$P'(x) = -60x^2 + 6\,000$	✓ derivative/ afgeleide A (1)
8.3	For maximum / Vir maksimum: $P'(x) = 0$ $-60x^2 + 6\,000 = 0$ $-60(x+10)(x-10) = 0$ OR/OF $x = \frac{-(0) \pm \sqrt{(0)^2 - 4(-60)(6\,000)}}{2(-60)}$ OR/OF $x^2 = \frac{6\,000}{60}$ $x = 10$ or/of $x \neq -10$ $P(10) = -20(10)^3 + 6\,000(10) - 10\,000$ $= R30\,000$	✓ equating derivative to/ stel afgeleide aan 0 A ✓ factors/faktore /SF CA ✓ correct value of x/ korrekte waarde vir x CA ✓SF CA ✓ Maximum Profit / maksimum wins CA NPU (5)
		[7]

QUESTION/VRAAG 9

9.1.1	$\int -\frac{6}{x} dx = -6 \ln x + C$	✓ $-6 \ln x$ A ✓ C A (2)
9.1.2	$\int (3x - 4)(x + 2) dx$ $= \int (3x^2 + 2x - 8) dx$ $= x^3 + x^2 - 8x + C$	✓ S A ✓ x^3 CA ✓ x^2 CA ✓ $-8x + C$ CA (4)
9.2.1	$\int 2^x dx = \frac{2^x}{\ln 2} + C$	✓ $\frac{2^x}{\ln 2} + C$ A (1)
9.2.2	$A = \int_{-2}^0 f(x) dx = \int_{-2}^0 2^x dx$ $= \left[\frac{2^x}{\ln 2} \right]_{-2}^0$ $= \left[\frac{2^0}{\ln 2} \right] - \left[\frac{2^{-2}}{\ln 2} \right] = \frac{3}{\ln 2}$ $= \frac{3}{\ln 2}$ OR/OF $\approx 1,08 \text{ units}^2 / \text{eenhede}^2$ $B = \int_2^3 f(x) dx = \int_2^3 2^x dx = \left[\frac{2^x}{\ln 2} \right]_2^3$ $= \left[\frac{2^3}{\ln 2} \right] - \left[\frac{2^2}{\ln 2} \right] = \frac{4}{\ln 2}$ $= \frac{4}{\ln 2}$ OR/OF $\approx 5,77 \text{ units}^2 / \text{eenhede}^2$ $\therefore \frac{B}{A} = \frac{4}{\ln 2} - \frac{3}{\ln 2} = \frac{16}{3}$ OR/OF $\therefore \frac{B}{A} = \frac{5,77}{1,08} = 5,34$ OR/OF	✓ Area notation using integrals/Area-notasie met gebruik van integrale M ✓ SF CA ✓ A value / waarde CA ✓ SF CA ✓ B value / waarde CA ✓ M CA

	$B - 4A = \frac{4}{\ln 2} - 4 \left(\frac{\frac{3}{4}}{\ln 2} \right) \quad \text{OR/OF} \approx 5,77 - 4(1,08)$ $= \frac{1}{\ln 2} \approx 1,45$ OR/OF $B > 4A \quad \frac{4}{\ln 2} > 4 \left(\frac{\frac{3}{4}}{\ln 2} \right) \quad \text{OR/OF} \quad 5,77 > 4(1,08)$ $\frac{4}{\ln 2} > \frac{3}{\ln 2} \quad 5,77 > 4,32$ ∴ The learner's claim is NOT VALID / Die leerder se bewering is NIE GELDIG NIE.	✓ Conclusion / Gevolgtrekking CA (7)
		[14]
	TOTAL/TOTAAL	[150]